

β -OPSD: derive the target with policy optimization, train it with self-distillation

A controllable path from the reference policy to privileged teacher guidance

1 Policy Optimization View

β -OPSD Family

KL-regularized objective

$$\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [R(y; x, c)] - \beta D_{\text{KL}}(\pi_{\theta}(\cdot | x) \parallel \pi_{\text{ref}}(\cdot | x))$$

Make β explicit and controllable

β controls how far the student moves



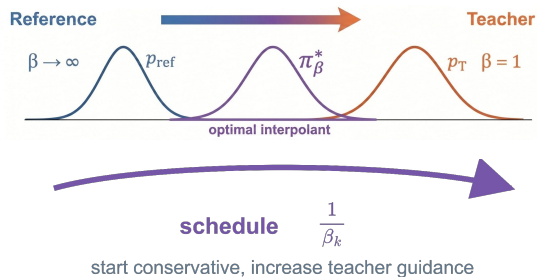
$\beta = 1 \Rightarrow$ vanilla OPSD
minimize reverse KL to teacher

2 Closed-Form Target

Each β schedules an optimal target

Closed-form optimal target

$$\pi_{\beta}^*(y | x, c) \propto \pi_{\text{ref}}(y | x)^{1 - \frac{1}{\beta}} p_T(y | x, c)^{\frac{1}{\beta}}$$



3 Efficient Training

Distill towards the RL-derived solution

Token-level logit interpolation

$$\tilde{z}_t = \left(1 - \frac{1}{\beta_k}\right) z_{\text{ref},t} + \frac{1}{\beta_k} z_{T,t}$$

Return-to-go credit

future target mismatch updates earlier tokens



Efficient distillation

no direct RL optimization

β -OPSD keeps the efficiency of OPSD while adding principled control, smoother targets, and sequence-level credit